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## **Well-Design Optimization Enables Drilling Performance Improvement in Unconventional Shale Gas Project, Canada**

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### **Abstract**

In North America shale gas projects, operators aim to extend the horizontal sections to boost single well production and reduce drilling and completion costs. However, as the horizontal section length increases, especially when the laterals exceed 3500 meters, numerous complex drilling challenges arise. These include issues with weight transfer during slide drilling, high downhole friction and torque, mud losses caused by high mud weight and friction loss in long horizontal section, short service lives and low efficiency of downhole tools, limited capability to track thin strata due to restricted logging equipment, and difficulties in deploying production casing, including risks such as differential sticking and prolonged running times.

To address these challenges, this paper presents an optimized well design strategy for the drilling of unconventional extended-reach unconventional horizontal wells. Firstly, a slim hole design is adopted to cut costs related to casing, cement, and drilling fluids. Secondly, the upper drilling string is switched from 4" to 4.5" drill pipes to provide enough torque and hydraulic power, and the Bottom Hole Assembly (BHA) utilizes even-wall stator positive displacement motors (PDM) and motorized rotary steerable systems (RSS) to enhance drilling efficiency in the intermediate and horizontal sections. Third, the oil-based mud (OBM) system is optimized for stability, low friction, and employs clay-free invert emulsion, complemented by managed pressure drilling (MPD) to control ECD. Then, downhole vibration sensors integrated into RSS enable predictive maintenance against tool fatigue. Next, drilling bit optimization based on formation strength analysis increases rate of penetration and extends bit life. Next, dual-telemetry xBolt (MWD) technology, combined with azimuthal gamma ray logging (LWD), which precisely reveals bed crossings and boundaries, optimizes wellbore placement. Finally, a specialized casing running tool and robust casing connection ensure successful deployment of 5.5-inch casings in long laterals.

These optimizations delivered outstanding outcomes. Several records were broken, such as the longest 4,875-meter horizontal slim-hole section drilled in a single bit run, with the average sweet zone drilling ratio of 96%, the longest single-run deployment of 5.5-inch and 4.5-inch hybrid casing at a depth of 8,157-meter,

and a rapid 17-day drilling cycle without incidents. In recent years, numerous horizontal wells have extended beyond 4,000m in lateral length, achieving an average horizontal rate of penetration (ROP) exceeding 50m/h and an average drilling duration of within 20 days.

Finally, These drilling optimizations deployed in the Duvernay shale project have notably reduced per-meter costs and enhanced drilling speed, setting a new benchmark within the Western Canadian unconventional project. These practices has set the stage for continued drilling of extended reach horizontal wells in the development of similar unconventional reservoirs.

## Introduction

The Duvernay Shale Gas Project is located in the western part of the Western Canadian Basin, Alberta, approximately 430 km northwest of Calgary. The main development formation is the Devonian Duvernay Shale, with an effective reservoir thickness of 30-45 meters. The Duvernay formation is further subdivided into sublayers A, B, C, and D from top to bottom, among which Sublayers Peak 3 to 5 of Sublayer D are vertical "geological sweet spots" with a thickness of approximately 3-5 meters.

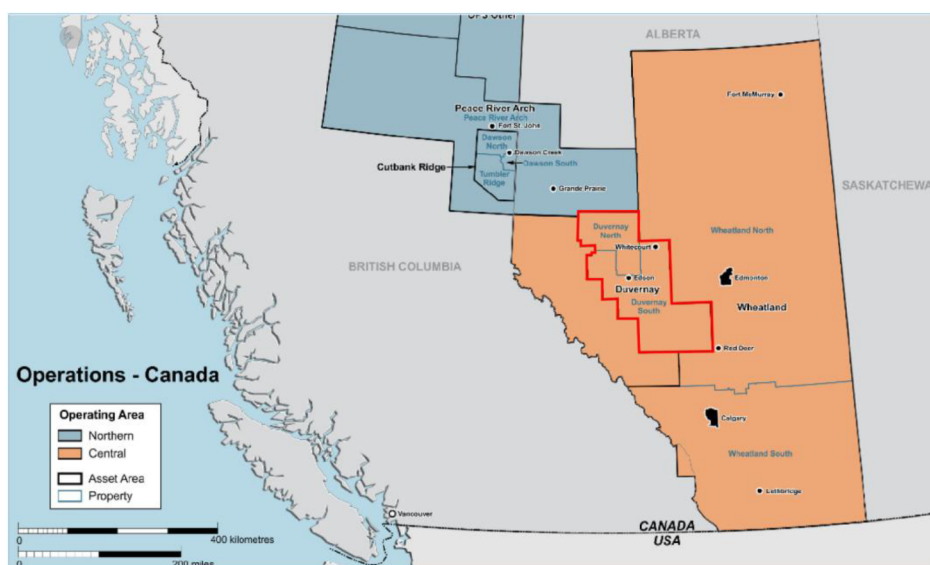


Figure 1—General location and field specific map.

Subsurface analysis places the Duvernay Formation between 3050m and 3800m depth, exhibiting an average pore pressure gradient (PG) of 1.80 kg/m<sup>3</sup>. While predicted fracture gradients typically reach 2.35 kg/m<sup>3</sup>, localized zones display reduced values down to 2.27 kg/m<sup>3</sup>. Its reservoir properties are characterized by low porosity (3.0%~6%) and extremely low permeability (0.0001~0.0003mD). Given such tight reservoir conditions, the development of shale gas projects in this formation necessitates the use of long horizontal laterals combined with large-scale hydraulic fracturing to achieve successful production.

In terms of stratigraphic composition, the upper interval (0~2200m) features alternating shales and sandstones; intermediate depths (2200~3200m), encompassing Banff and Ireton strata, comprise compact carbonates like limestone/dolomite; Below 3200m, the Duvernay Formation itself is composed of tight shale. Pressure systems vary with depth: the upper formations (0~3050m) fall under a normal pressure regime, with pressure coefficients ranging from 1.00 to 1.10. Conversely, lower sections including Ireton's base and entire Duvernay show elevated pressures climbing from 1.25 to 1.80, though lost circulation risks arise in Duvernay due to natural fractures.

Due to the extended-reach nature of the Extended Reach Drilling (ERD) well, compounded by the narrow pore pressure and fracture pressure window, long horizontal slim-hole drilling face both mechanical and hydraulic challenges. Key challenges include:

Due to the extended reach nature of the long and slim hole

1. **Weight transfer difficulties** caused by poor wellbore cleaning and high friction factor during drilling.
2. **Excessive mud losses** resulting from inappropriate drilling fluid property and ECD limitation.
3. **Frequent tool failures and signal loss** due to high downhole vibrations
4. **Inadequate data transmission**—MWD (Measurement While Drilling) and LWD (Logging While Drilling) systems lack sufficient bandwidth and transmission rates to support high-speed drilling operations.
5. **Casing deployment issues**—differential sticking and buckling risks making it difficult to deploy production casing in a single run.

To address these challenges and further enhance the overall drilling efficiency and safety of ultra-long horizontal wells, offset wells were first analyzed, and lessons learned from years of horizontal well execution were integrated into the process; additionally, a multi-functional team was involved to collaborate on aligning with a fit-for-purpose well design.

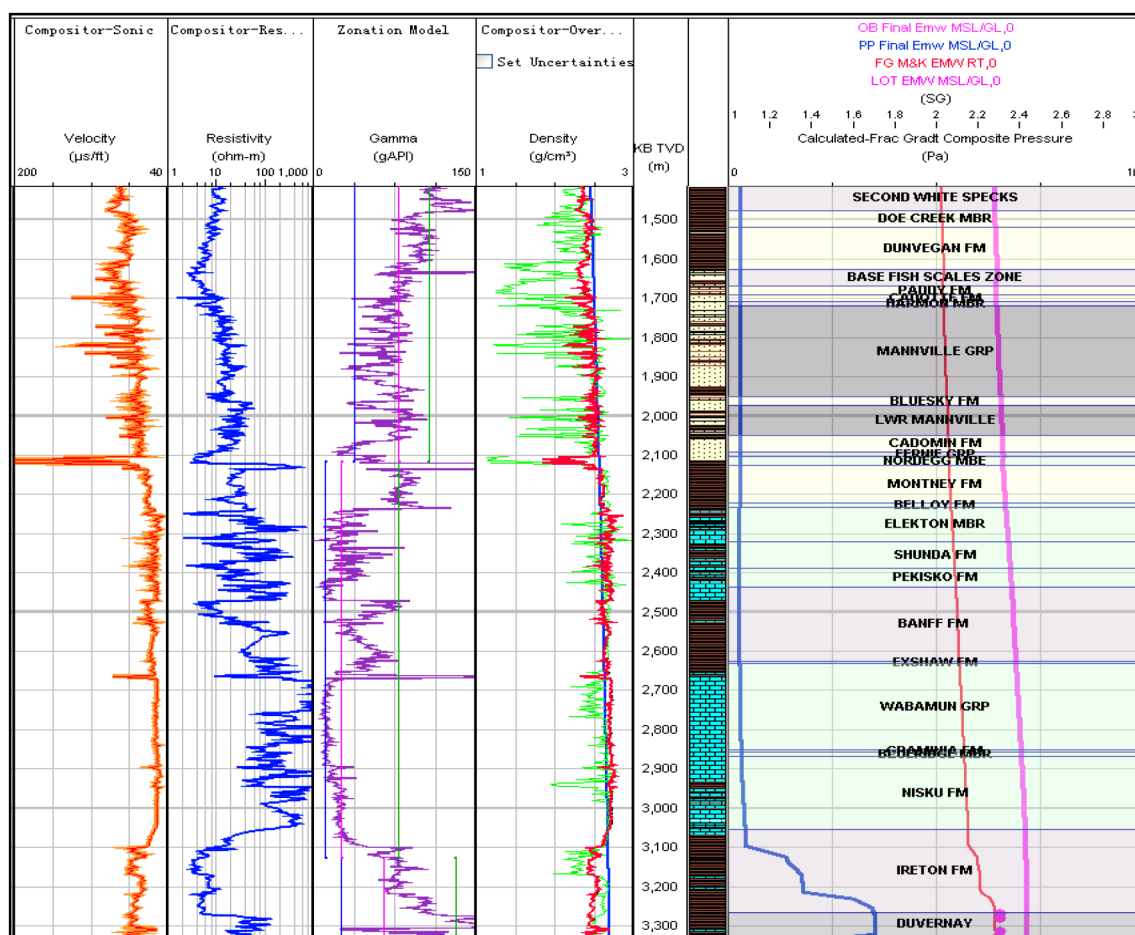


Figure 2—Stratigraphic column with formation pressure gradient.

## Slim Hole Casing Design

Standard producer well designs in North American shale fields incorporate zonal isolation in every hole section above the pay zone for enhanced mechanical stability. The typical design includes: A 17 1/2-in surface hole is vertically drilled with a spud mud then 13 3/8-in, 81.1 kg/m, J-55, BTC casing is run as / conductor to protect shallow aquifers. A 12 1/4-in hole is then drilled using oil based mud (OBM) by building inclination to 30-40° then running the 9 5/8-in, 64.7 kg/m, L-80, LTC casing into very competent Ireton shale to isolate the upper loss of circulation and potentially sour aquifer zones. Below this casing a 8 1/2-in hole is drilled with oil based mud (OBM) across low permeable and unstable layers of shale, while continuously bringing the inclination up-to 88° ~90° at the target entry at ±3,260 m-TVD, the 8 1/2-in hole is drilled with OBM and completed the lateral sections, then running the 5 1/2-in, 44.2 kg/m, P-110, casing

Based on integrated assessment of technical feasibility and operational effects, a purpose-designed slim-well casing strategy was implemented for Duvernay Shale operations, representative examples shown in Fig. 3. Typical execution starts with a 13 3/4-inch pilot hole and 9 5/8-inch J-55 surface casing (53.57 kg/m @ 618m MD), progresses via an 8 3/4-inch directional section cased with 7 5/8-inch L-80 ICY pipe (44.20 kg/m; set ~15-20m TVD in Ireton FM), and continues building turn through lower Ireton/upper Duvernay using a 6 3/4-inch assembly at 5-7° DLS towards a near-horizontal landing zone (88.0°-90.0°). Against conventional step-out designs incorporating 3800m laterals, this optimized schema offers major savings: casing steel intake falls 35% (588 tons -> 383 tons/well). This reflects shifting from heavier weights (prev.: 13 3/8" @ 81.1kg/m / 9 5/8" @ 64.7kg/m / 5 1/2" @ 44.2kg/m) to lighter counterparts (new: 9 5/8" @ 53.57kg/m / 7 5/8" @ 44.2kg/m / 5 1/2" @ 34.2kg/m or 4 1/2" @ 22.3kg/m). Secondary benefits encompass a 37% decline in annular cement requirements and a 36% smaller total hole volume, diminishing mud needs, disposal costs, and losses from oil carryover on cuttings.

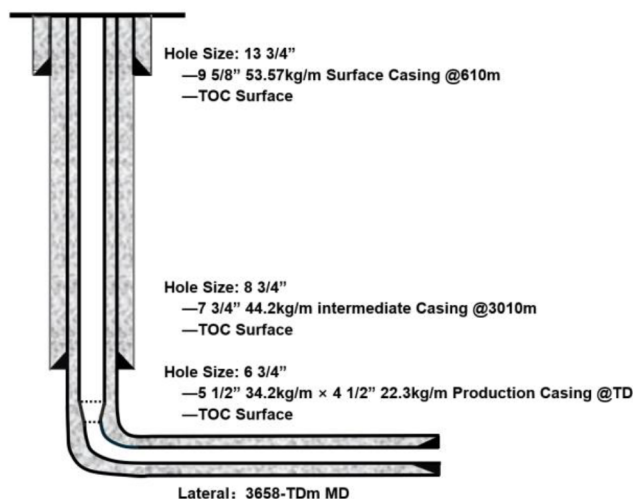


Figure 3—Well profile of Slimhole casing design.

While traditionally viewed as inferior because of low ROP, tool reliability issues, high ECD, and pipe-handling friction, slim well technology has advanced considerably. Modern solutions—utilizing purpose-built tight-clearance pipes, high-output motors, and optimized bits—are augmented by advanced monitoring (MWD/LWD), suppressive oil-based fluids, and MPD. In-depth modeling of the entire drill string's mechanics (buckling, vibration) and hydraulics guides critical decisions on BHA design, hole size, and formation placement. This systematic approach resolves historical limitations, enabling slim-hole sections to match or surpass the performance of larger, conventional wellbores.



## Trajectory Design and Torque & Drag Analysis

Drilling platform typically host 8~16 wells organized into double rows, spaced 40~50 meters apart with 10-meter intervals between adjacent wellheads. To mitigate the waste of dead gas zone below the structure, these wellbores utilize a specialized 'spoon-shaped' profile composed of several distinct segments "vertical-build-hold-turn-build-hold-build-horizontal". Initially, a vertical hole is drilled to roughly  $\pm 650$  meters. The second phase begins at 650 meters, starting a gradual inclination build at a shallow angle to reach  $10\sim 20^\circ$ . This trajectory, designed with a low Dogleg Severity (DLS) of  $1\sim 3^\circ/30\text{m}$ , navigates through the Nisku formation. As this phase progresses, both the inclination and azimuth are continuously adjusted to align with the final horizontal target. A technical casing string is set at the top of the Ireton formation, with this section adhering to a stricter DLS limit of  $3\sim 4^\circ/30\text{m}$ . The third and final phase initiates from the casing shoe at an initial inclination of  $27^\circ\sim 40^\circ$ . From this point, the well path steepens, increasing inclination to  $90^\circ$  to arrive at Target Point B. An intermediate kick-off point, located 50~100 meters short of the target, features a controlled DLS of  $3\sim 6^\circ/30\text{m}$  for the final approach.

Table 1—Standard horizontal well path design.

| Measured Depth (m) | Inclination ( $^\circ$ ) | Azimuth ( $^\circ$ ) | Vertical Depth (m) | +N/S (m) | +E/-W (m) | Dogleg ( $^\circ/30\text{m}$ ) | Build Rate ( $^\circ/30\text{m}$ ) | Turn Rate ( $^\circ/30\text{m}$ ) |
|--------------------|--------------------------|----------------------|--------------------|----------|-----------|--------------------------------|------------------------------------|-----------------------------------|
| 0.00               | 0.00                     | 0.00                 | 0.00               | 0.00     | 0.00      | 0.000                          | 0.000                              | 0.000                             |
| 650.0              | 0.00                     | 0.00                 | 650.0              | 0.00     | 0.00      | 0.000                          | 0.000                              | 0.000                             |
| 880.0              | 11.5                     | 75.92                | 878.46             | 5.6      | 22.31     | 1.500                          | 1.500                              | 0.000                             |
| 2778.0             | 11.5                     | 75.92                | 2738.36            | 97.69    | 389.34    | 0.000                          | 0.000                              | 0.000                             |
| 2999.97            | 11.5                     | 319.82               | 2956.03            | 132.26   | 400.5     | 0.000                          | 0.000                              | -15.57                            |
| 3029.00            | 14.5                     | 319.82               | 2984.31            | 137.25   | 396.29    | 3.100                          | 3.100                              | 0.000                             |
| 3088.00            | 14.5                     | 319.82               | 3041.43            | 148.54   | 386.76    | 0.000                          | 0.000                              | 0.000                             |
| 3292.34            | 35.5                     | 319.82               | 3224.85            | 215.47   | 330.24    | 3.500                          | 3.500                              | 0.000                             |
| 3654.80            | 89.93                    | 319.82               | 3384.84            | 452.42   | 130.13    | 4.510                          | 4.510                              | 0.000                             |
| 7080.00            | 89.93                    | 319.82               | 3385.56            | 3069.35  | -2079.64  | 0.000                          | 0.000                              | 0.000                             |

Trajectory control in the vertical and curve-building sections relies on a conventional PDM motor monitored by MWD tools. For the horizontal displacement, an even-wall thickness motor provides steering capability alongside MWD feedback. Drag and torque analysis shows When the friction coefficient of the open hole section exceeds 0.4, the maximum extension length of the horizontal well under sliding mode shall not exceed 6,000m for weight transfer and torque issue.(Block weight: 22 kdaN, Cased Hole FF: 0.25, Open Hole FF: 0.2 to 0.5). Addressing the recurring challenge of differential sticking induced by high contact forces, operators began implementing Rotary Steerable Systems (RSS) after 2018. The adoption of RSS led to marked improvements: improved borehole quality, superior cuttings removal, decreased trouble time complexity, and achievement of a record horizontal section reaching 4,400m.

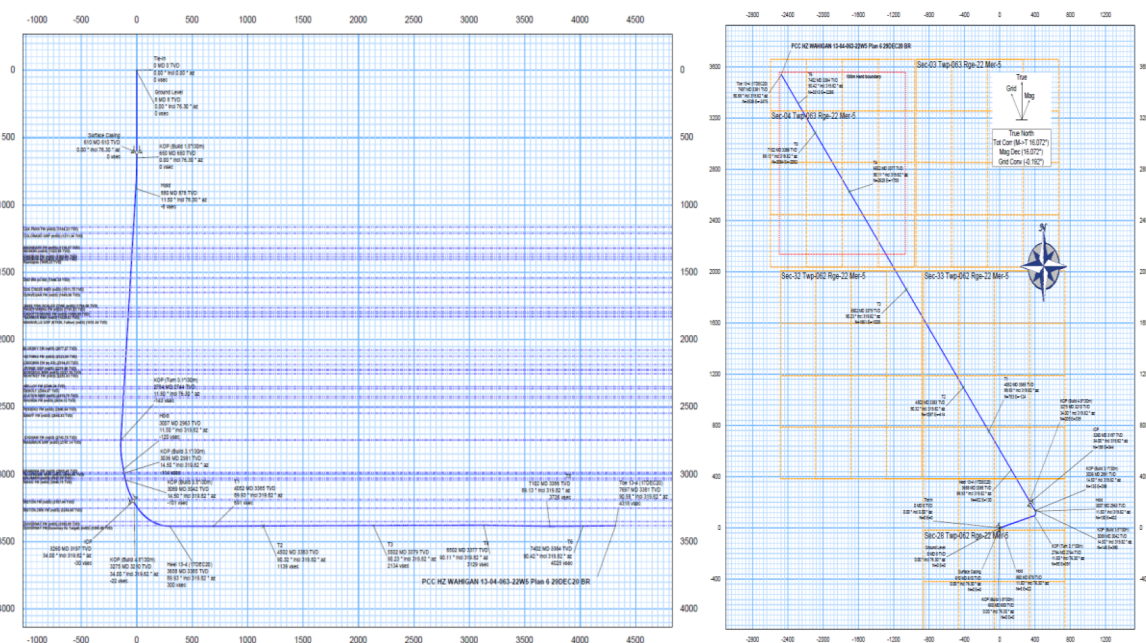


Figure 4—Typical horizontal well trajectory of Duvernay project.

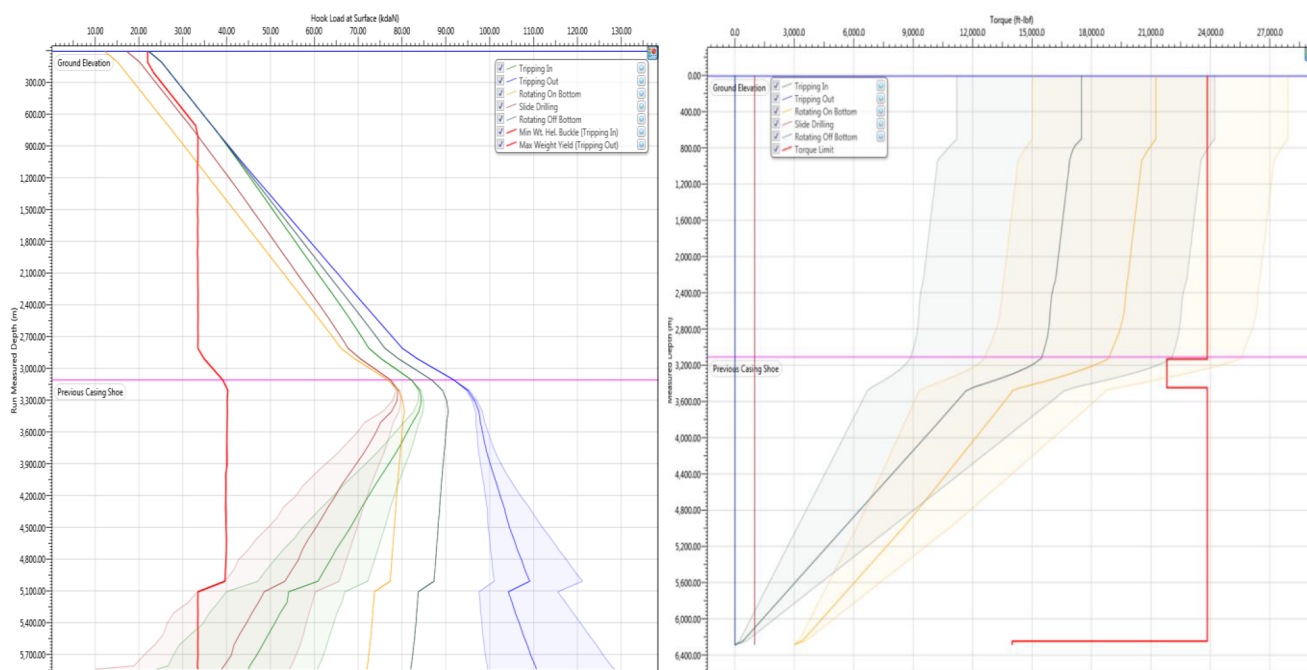


Figure 5—Drag and torque analysis for 6 3/4-inch horizontal section. Open Hole FF range between 0.2 to 0.5

## Drilling Pipe and Bottom Hole Assembly

### Drilling Pipe optimization

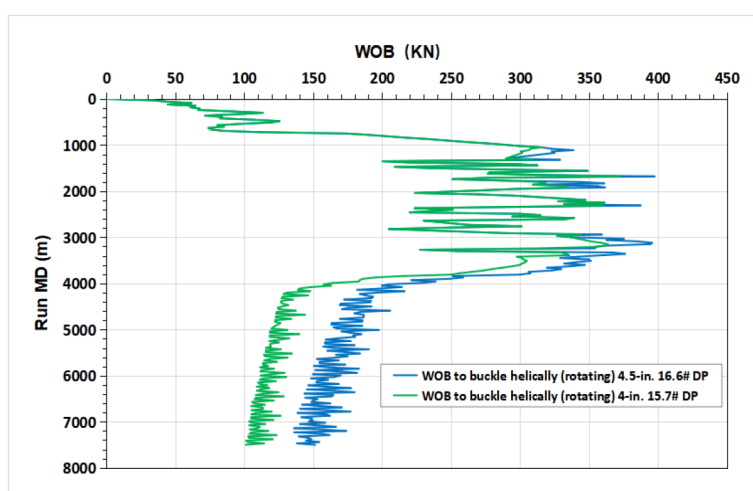
Before 2016, project plans for drilling 6 $\frac{1}{8}$ -inch slim horizontal sections included two potential solutions—using either 4-inch/NC40 tubulars (5 $\frac{1}{4}$ -inch OD) or thicker 4.5-inch/NC46 variants (6 $\frac{1}{4}$ -inch OD). The latter proved unfeasible due to its need for a minimum clearance opening of 7 $\frac{7}{8}$ -inch, conflicting with existing well designs. Advancements in slim-hole technology have since enabled modern 4.5-inch high-level pipes equipped with streamlined CET43 connectors (5 $\frac{3}{8}$ -inch OD). These components offer significant advantages over previous systems, including enhanced maneuverability in confined boreholes, increased

stability against compressional buckling, improved torque transmission, more effective load distribution during drilling, and extended service life by mitigating fatigue cracking.

**Table 2—traditional API DP vs High-level connection DP**

| Specification                         | API DP         |               | High-level DP |
|---------------------------------------|----------------|---------------|---------------|
| Pipe body Diameter(in)                | 4" × 3.826"    | 4.5" × 3.826" | 4.5" × 3.826" |
| Pipe Weight(ppf)                      | 15.7#          | 16.6#         | 16.6#         |
| Pipe body Torsional Strength(ft·lbf)  | 36,134         | 43,450        | 44,362        |
| Connection Type                       | NC40           | NC46          | CET43         |
| Connection Diameter(in)               | 5.25" × 2.563" | 6.25" × 3.0"  | 5.375" × 3.0" |
| Max Make up torque(ft·lbf)            | 16,651         | 23,400        | 31,700        |
| Connection Torsional Strength(ft·lbf) | 27,752         | 39,000        | 48,800        |

A helical-buckling analysis was conducted for rotary drilling operations in the 6.75-inch hole section. Specifically, when drilling on bottom at the total depth (7500 m MD), helical buckling initiates at a weight on bit (WOB) of 11.4 tons when using 4-inch drillpipe (weight class: 15.7 lb/ft). In contrast, when deploying 4.5-inch drillpipe (weight class: 16.6 lb/ft) under the same downhole conditions, the onset of helical buckling occurs at a higher WOB of 14.8 tons. In extended-reach laterals measuring 3800 meters, upgrading from 4-inch to 4.5-inch drillpipe yields measurable performance gains: the heavier 4.5-inch drill pipe tolerates an extra 3.4 tons of WOB prior to helical buckling initiation. This upgrade increases sustainable WOB capacity throughout the entire lateral section by 10~25%. Figure 6 presents the simulation results, illustrating the minimum allowable WOB (to prevent helical buckling) at various positions along the drillstring during on-bottom rotation stage. In most operational scenarios, increasing the applied WOB leads to a proportional improvement in the rate of penetration (ROP).



**Figure 6—Helical-buckling analysis with 4-in and 4.5-in. drillpipe.**

### Intermediate BHA Optimization

Before 2015, the standard practice for drilling the 8.75-inch intermediate section in Duvernay plays was the deployment of a 6.75-inch hard rubber motor. Its application was initially successful for excavating the relatively unconsolidated Cretaceous strata of sandstone and mudstone. However, upon encountering the exceptionally durable limestone formation capping the Duvernay shale, the existing motor's power output proved insufficient. Consequently, crews had to perform a time-consuming tripping operation to replace the

BHA with a more suitable motor capable of finishing the section to the ICP. On average, this multi-step process consumed approximately ten days and required two to three separate bit runs per well.

To overcome the limitations of traditional motors, a technologically superior 7.25-inch high-torque, low-speed motor was chosen for drilling the Duvernay intermediate sections. Compared to standard motors, this engineered solution offers vastly improved performance: sustained bit torque increased by 72%, and resistance to differential pressure surged by 64%, shows in Fig. 7. These attributes directly translate to longer on-bottom drilling time and significantly fewer bit trips, critical factors in high-wear environments. Field validation came immediately, with a standout performance of 2,658 meters drilled continuously in one run—accomplished incident-free within six days, setting a benchmark. Practical application from 2020-2024 demonstrated its reliability, achieving single-run completion of the intermediate section on 14 of the 20 wells drilled.

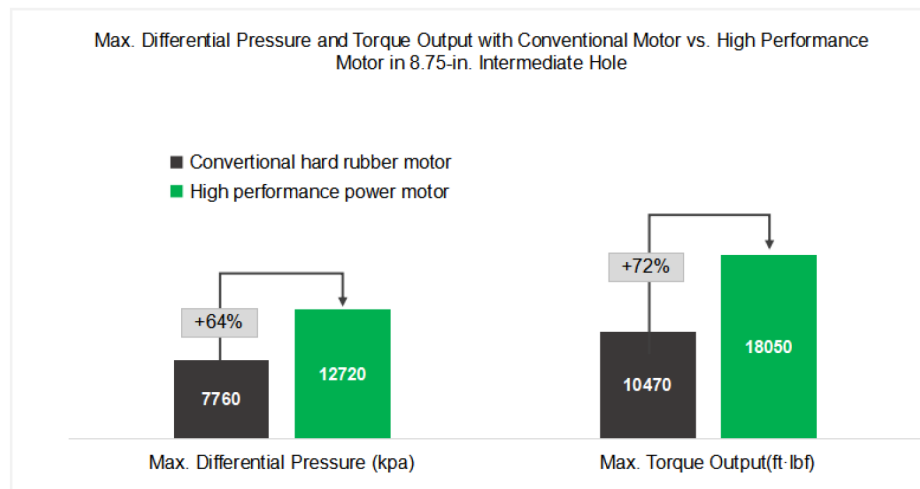


Figure 7—Conventional motor vs high-performance motor

### Motorized RSS BHA

In conventional well drilling for Duvernay's 6.75-inch deviated and lateral intervals follows a two-stage process. First, a mud motor drives the progression from the Intermediate Casing Point (ICP) through the build section, navigating inclinations from 40 to 90 degrees to reach the designated landing point. Subsequently, control shifts to a Rotary Steerable System (RSS) integrated with an M/LWD assembly, which guides the well path from the landing point to the final Target Total Depth (TD). Despite its prevalence, this standard workflow presents several operational challenges.

1. **High dogleg severity (DLS)** and reduced rate of penetration (ROP) during the slide-drilling phase.
2. **Need for at least two separately bottom hole assemblies (BHAs)** to reach the planned TD.
3. **Increased costs** due to the use of two polycrystalline diamond compact (PDC) bits—one for the motor run and another for the RSS operation.

To resolve these drilling challenges, a high output Motorized Rotary Steerable System (Motorized RSS) was added to the BHA. This system represents a key technological shift, uniting a mud motor and an RSS in a single, cohesive assembly. The primary benefit is sustained rotation throughout the drilling section, facilitating smooth, drag-free steering. Moreover, the motor enhances drilling efficiency by supplying additional torque and speed to the bit while acting as a dampener for damaging torsional vibrations. This protective measure reduces the alternating stress on the remainder of the drill string, effectively cutting down on fatigue-related wear.

Table 3—Down hole tool specification.

| Tool Name                           | Mud Motor      | Mud Motor      | Mud Motor   | RSS         |
|-------------------------------------|----------------|----------------|-------------|-------------|
| Well section                        | 13 3/4"        | 8 3/4"         | 6 3/4"      |             |
| Series                              | Dyna Drill 962 | Dyna Drill 725 | Claw650(XF) | PD Orbit475 |
| Size (inch)                         | 9.625"         | 7.25"          | 5.25"       | 4.75"       |
| Lobe                                | 3/4            | 6/7            | 5/6         | -           |
| Stages                              | 6              | 8.2            | 8.3         | -           |
| Flow rate range (gpm)               | 600~1200       | 500~750        | 150~400     | 120~355     |
| Speed Range (rpm)                   | 132~270        | 130~200        | 260         | 220~350     |
| Rated Diff Press (psi)              | 900            | 1850           | 2328        | 2000        |
| Rated torque (ft-lbf)               | 11480          | 18050          | 8984        | 9000        |
| Rated power (HP)                    | 579            | 628            | 445         | -           |
| Maximum temperature resistance (°C) | 120°C~130°C    | 120°C~130°C    | 120°C~130°C | 150°        |

The performance improvements achieved with the motorized RSS over conventional drive systems were significant. Operating at a pump rate of 300 gallons per minute (gpm) and a downhole weight on bit (WOB) of 14 tons, the average ROP in the 6.75-inch hole section exceeded 53 meters per hour—representing a 83% increase (as illustrated in Fig. 8). Moreover, stick-slip vibrations were reduced by 42% as anticipated, as illustrated in Fig. 9.

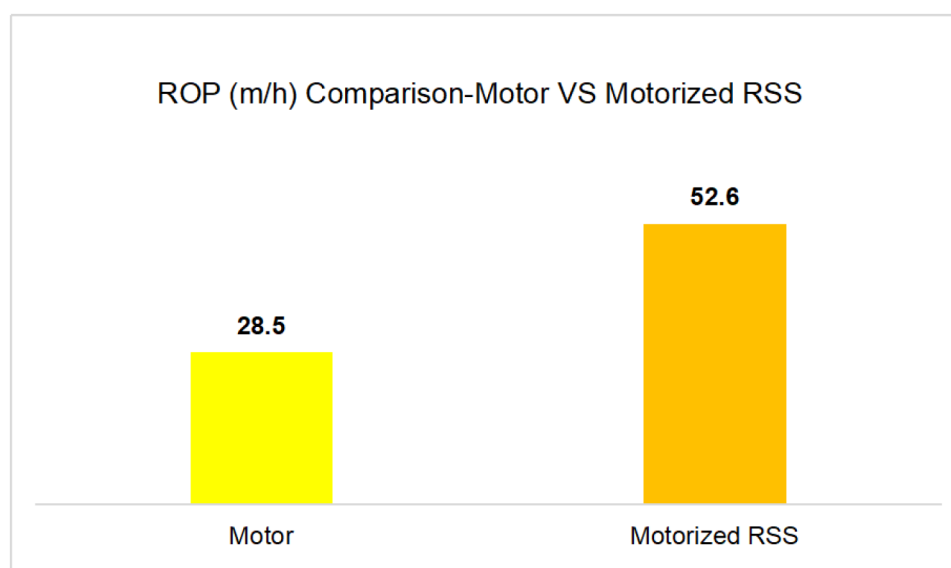


Figure 8—ROP- Motor vs Motorized RSS BHA.



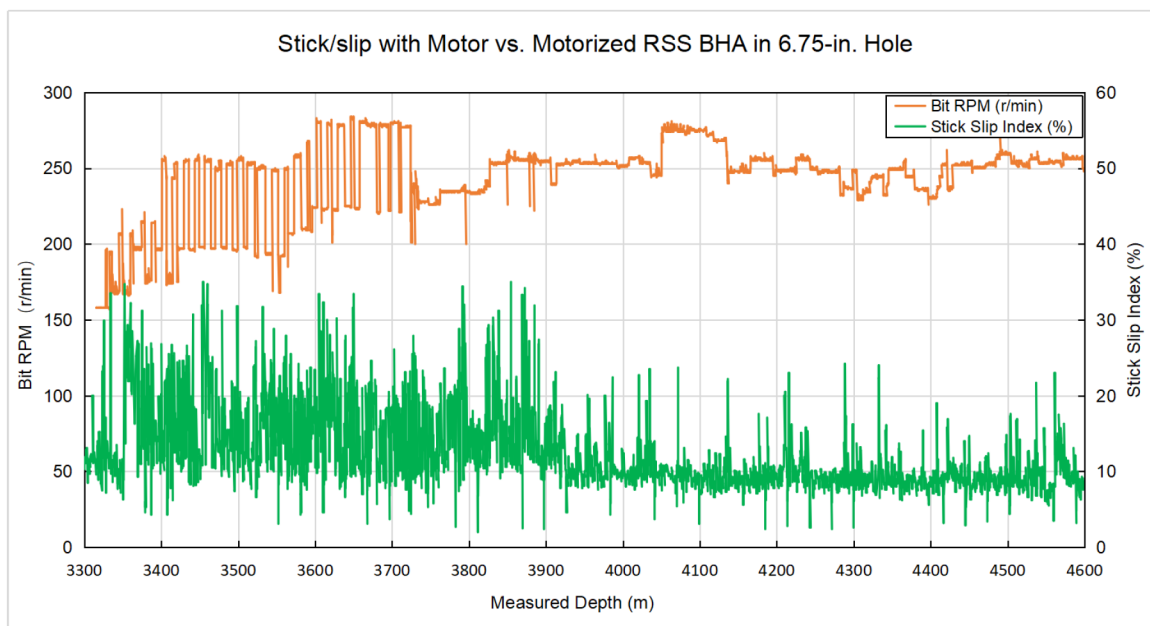


Figure 9—Stick/slip index- Motor vs. Motorized RSS BHA.

## Managing Drilling Fluid and ECD

Slim hole drilling using higher pipe-to-hole ratios (e.g., 4.5-inch drillpipe in 6.75-inch hole) boosts performance through improved buckling resistance, torsional stiffness, lower hydraulic friction losses, and faster annular flow for effective horizontal cleaning. Yet this setup raises significant challenges: elevated annular friction during pumping spikes equivalent circulating density (ECD), risking fluid loss at target flow rates. Lightening mud weight to compensate increases instability danger during pump stops (e.g., connection times), as reduced equivalent static density (ESD) may cause hole collapse. While high ECD rarely threatens microdarcy unconventional, it remains critical in highly pressured, naturally fractured shales—particularly when expensive oil-based fluids are used.

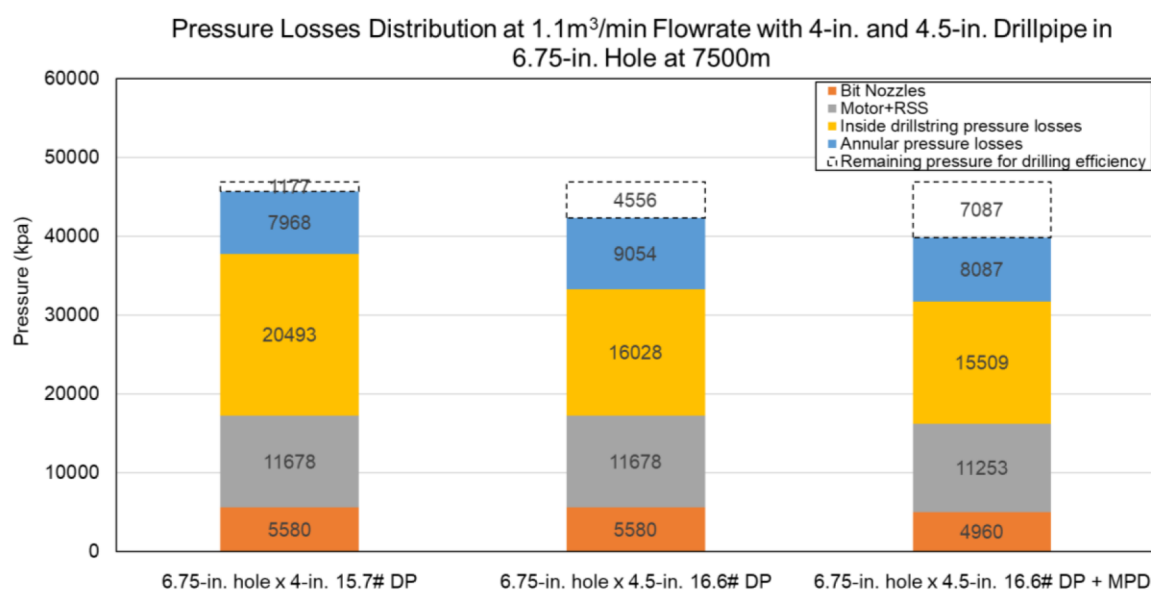
### High-Performance Oil Base Mud System

The Duvernay Shale's mineralogy—30~40% clay including reactive montmorillonite and illite—expands when exposed to water, posing serious drilling challenges. Consequently, operators in this play universally adopt oil-based muds for their superior wellbore stabilization capabilities. For extended 6.75-inch horizontal laterals, fluid designs must simultaneously combat shale swelling, mitigate lost returns, and minimize torque/drag. To address these needs, BaraXcel™ high-performance invert emulsion fluid was formulated for full-hole application across directional and lateral intervals. Its low-solids formulation delivers excellent shear-dependent rheology: preventing barite settling during circulation breaks while rapidly breaking gel strength upon restart—avoiding pressure surges that elevate equivalent circulating density (ECD) and reduce loss risks during pipe trips. EZ-MUL® NT emulsifier ensures robust phase stability. A synergistic blend of Baracarb™ ultrafine  $\text{CaCO}_3$  particles and Barofibre™ fibrous additives seals microfractures, forming an ultrathin, impermeable filter cake that precisely controls invasion into pressure-sensitive formations.

### ECD management

At standard drilling flow rates above  $1.0 \text{ m}^3/\text{min}$ , switching from 4-inch to 4.5-inch drillpipe leverages its larger internal diameter to substantially cut pump pressure requirements. This reduces hydraulic friction losses, freeing more energy for efficient drilling while boosting annular velocity to enhance hole cleaning in horizontal intervals. Calculations comparing premium-connected 4-inch (15.7#) and 4.5-inch (16.6#) pipes

at 1.1 m<sup>3</sup>/min reveal total system losses dropping from 28.5 MPa to 25.1 MPa—a 3.4 MPa savings (see Fig. 10). With BHA components (RSS/motor/bit) consuming 17.3 MPa, the remaining usable headroom on a 52 MPa pump system (with a 46.9 MPa relief valve setting) jumps from 1.2 MPa for the smaller string to 4.5 MPa for the larger one. This extra margin directly translates into increased available hydraulic power for optimizing ROP through higher motor differential pressure, HSI, or supplemental flow allocation.



**Figure 10—Pressure Losses Distribution at 1.1m<sup>3</sup>/min Flowrate with 4-in. and 4.5-in. Drillpipe.**

To bridge the gap between equivalent static density (ESD) and circulating density (ECD), operators increasingly turn to managed pressure drilling (MPD). Among its variants, applying surface backpressure via rotating control devices and chokes is now standard practice for stabilizing ECD in unconventional wells. This method counteracts annular pressure decline during pump stops (like pipe connections), preserving consistent bottomhole pressure. In Alberta's Duvernay Shale, this MPD technique cuts costly downtime by preventing cycles of wellbore breathing, fluid loss, and kicks—common failure modes in challenging formations. Field data below demonstrate its effectiveness during main hole drilling with a 1560 kg/m<sup>3</sup> mud system at 1.1 m<sup>3</sup>/min flow rate.

Managed Pressure Drilling (MPD) enables ultra-slender well designs by reducing ECD, directly enhancing rate of penetration (ROP). Lower fluid densities decrease confining pressure at the bit, breaking rock more readily. Particularly beneficial in slim boreholes, near-balanced MPD—matching bottomhole pressure to formation pore pressure—accelerates ROP through dual mechanisms: weakened rock response and improved cuttings release from the formation face.

## Drilling Parameters Enhancement

Drilling performance was optimized through parameter adjustments in a 6<sup>7</sup>/<sub>8</sub>" horizontal section: flow rate rose from 0.9 to 1.2 m<sup>3</sup>/min, pump pressure jumped from 28 MPa to 40 MPa, WOB doubled from 8 to 16 tons, and RPM increased from 50 to 100. This transformed ROP surged from 15 m/h to 50 m/h by boosting annular velocity and hole cleaning efficiency—critical for safe, high-speed drilling.

Table 4—Comparison of drilling parameters after enhancement.

| Indicator                           | Before             | After          |
|-------------------------------------|--------------------|----------------|
| Flow rate ( $m^3/min$ )             | 240                | 320            |
| Maximum Standpipe Pressure (Mpa)    | 28                 | 40             |
| Weight on Bit (Ton)                 | 8-9                | 13-16          |
| Surface RPM (r/min)                 | 80                 | 100            |
| Top Drive + Motor RPM (r/min)       | 150-250            | 250-350        |
| Drilling fluid density ( $g/cm^3$ ) | Non MPD, 1.75~1.85 | MPD, 1.55~1.65 |

Mitigating Downhole Vibration

Field trials confirmed drillstring harmonics as significant contributors to component fatigue (Greenwood 2016). To combat this in Duvernay wells, engineers deployed an RSS-mounted near-bit sensor package with accelerometers that transmits real-time torsional/lateral/axial vibration data via pulse signals. When hazardous levels occur, surface software flags specific modes and recommends fixes like RPM or WOB changes—guiding engineers toward optimal settings. Persistent issues trigger post-run analyses of recorded downhole data, enabling solutions through BHA redesign, bit swaps, or stabilizer additions. Figure 11 reveals severe shock events correlated with low-gamma intervals in 6.75" laterals; here, immediate RPM/WOB adjustments shielded sensitive tools like RSS and MWD systems from damage.

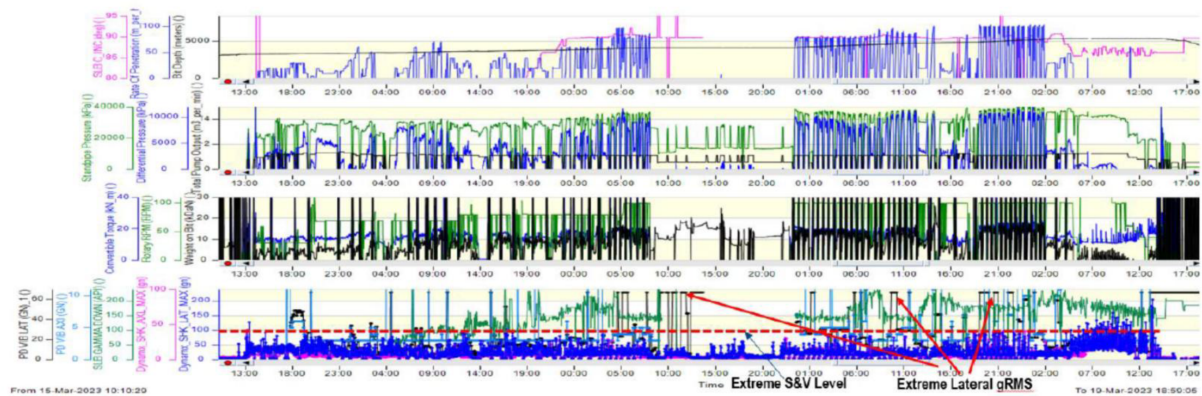


Figure 11—Mitigating downhole vibration by adjusting RPM and WOB.

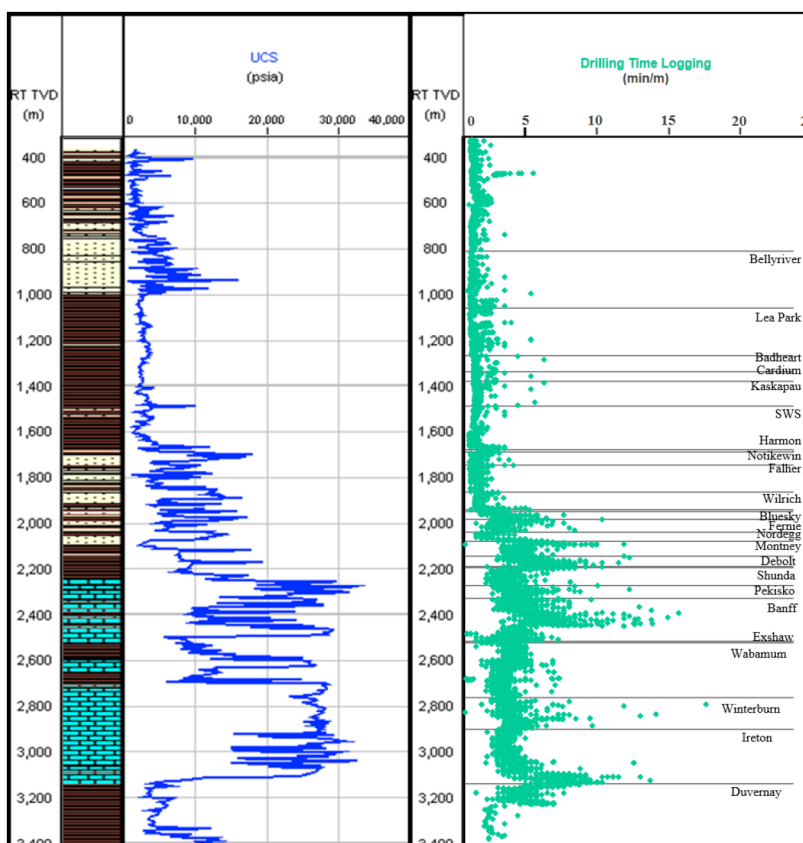


Figure 12—Rock strength and drilling time varies with depth.

## Bit Optimization

### Unconfined Compression Strength Analysis

Formation strength testing guides optimal bit selection by mapping rock resistance to compressional failure. Stratigraphic shifts occur above Lea Park (60~70% sand/silt), transitioning downward into shale-dominated sections (80~90%) lasting until Cardium sandstone appears with unconfined compression strength (UCS) values between 5~15 kpsi. Post-Base Fish Scales, sandstone exceeds 50% at higher gradients of 15~20 kpsi, where hard chert nodules create abrasive intervals causing rapid PDC bit damage—forcing reduced RPM and frequent replacements. Lower intermediate 8<sup>3</sup>/<sub>4</sub>" holes encounter severe lithological mixing (sandstone/dolomite/limestone) from Bluesky through Shunda, amplifying vibration effects and uneven cutter loading. Montney's abrasive siltstone accelerates wear at 20~30 kpsi, demanding durable cutters optimized for inefficient engagement scenarios. High-UCS zones in Pekisko, Banff, and Wabamun (20~40 kpsi) require aggressive yet tough bits balancing penetration speed and longevity. Finally, the heterogeneous 6<sup>3</sup>/<sub>4</sub>" section below Ireton marker into Duvernay combines high-quartz shales (>45%, UCS 10~15 kpsi) with 40–47° internal friction angles, necessitating wear-resistant designs prioritizing stability.

### Bit Selection

Bit selection entails defining foundational bit models and IADC codes through integration of well logs, UCS, and neighboring well performance histories. Engineering specifics encompass choosing structural materials, number of blades, aerodynamic contours, insert dimensions, jet orientation, gauge configurations, plus supplementary protection features like backup cutters.

**Surface section.** This interval exhibits low UCS below 10 kpsi alongside moderate abrasiveness. A 13<sup>3</sup>/<sub>4</sub>" diameter steel-bodied PDC bit with five blades carrying 19mm cutters is deployed here. Performance

enhancements include truncated gauge lengthening, 10~15° reverse rake geometry, plus compact parabolic blade shaping—all optimizing penetration rates.

**Intermediate section.** Covering 618m–3100m depth, this phase encounters unstable lithology shifts, variable UCS readings, elevated abrasion potential, plus directional steering demands post-initial vertical drilling. The prescribed assembly uses an six-blade matrix-body PDC (IADC M434) fitted with 13mm cutters. Design elements include backup cutting elements set at 20–30° negative rake angles restricting depth-of-cut for improved thermal management in hard/abrasive zones—boosting durability. Elongated gauge profiles enhance course correction capability, complemented by abrasion-resistant, heat-treated cutter alloys increasing both efficiency and service life.

**Horizontal section.** Within Duvernay's reduced UCS environment versus overburden layers, either five or four-blade matrix PDC bits (IADC M132) equipped with 13mm cutters are preferred. Medium parabolic blade shapes paired with shallow cone angles improve maneuverability for required dogleg severity compliance while accelerating hole cleaning action. Helical gauge arrangements maximize perimeter contact points, lowering steering friction and torque loads to elevate ROP and component longevity. Spherical backup inserts modulate per-revolution chip thickness, effectively dampening stick-slip oscillations.

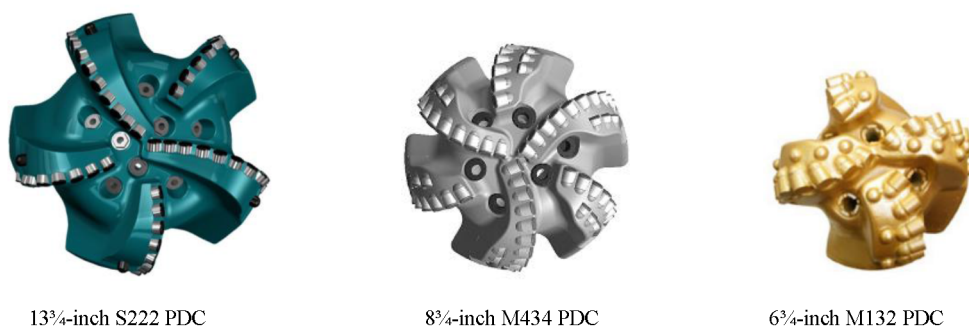


Figure 13—Bit design according to formation hardness.

## High-speed MWD technologies

One of the most pressing issues facing modern shale drillers is moving massive amounts of measurement-while-drilling (MWD), logging-while-drilling (LWD), and seismic vibration data uphole efficiently. Older single-mode Mud Pulse (MP) systems capped at <2 bps make it nearly impossible to boost ROP effectively. Recent innovations address this gap using dual-mode telemetry—merging both electromagnetic and pressure-wave transmission—allowing real-time communications without stopping progress in challenging horizontal sections typical of unconventional plays

To maintaining horizontal placement within the ideal reservoir zone, operators employ an integrated solution pairing azimuthal gamma(AGR) imaging with dual-telemetry xBolt MWD systems. This combination precisely maps thin Peak 3–5 layers (2~3m thick) and stratigraphic boundaries, empowering real-time geosteering adjustments backed by updated subsurface models. During a successful three-mile lateral run, high-fidelity AGR and RSS data streamed via xBolt's electromagnetic pulses at 12 BPS—using a downhole antenna receiver even at total depth. Duvernay fieldwork reveals actionable links between GR responses, penetration rates, and vibration levels; thus, ROP/vibration metrics supplement geosteering strategies, while added curve metrics refine interpretation accuracy.



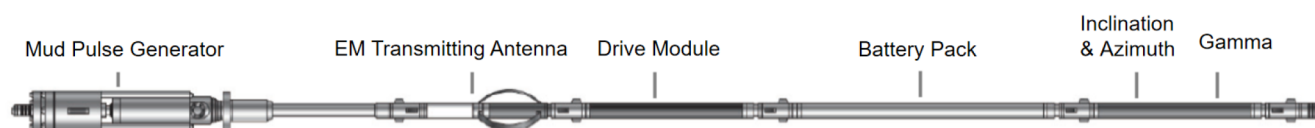


Figure 14—Diagram of the xBolt MWD instrument

## Casing Running Tool

The conventional casing running process is labor-intensive and involves significant human interaction with the casing. A Casing Running Tool (CRT) is a specialized device integrated with top drive systems, combining mechanical and hydraulic functionalities to facilitate casing running operations. Beyond enabling automated casing string connections, CRTs provide capabilities for rotating and lowering casing while allowing drilling fluid circulation. Meanwhile, CRTs reduce the number of personnel required for casing operations, offering dual advantages of enhanced operation efficiency and safety.

This casing-running system has been deployed over 50 horizontal wells within the Duvernay project. The external-grip CRT systems not only enables rapid rig-up (typically 15~25 minutes, compared to 2~3 hours with conventional equipment) for substantial rig time savings, but also reduces casing-wellbore friction by nearly 50% and an overall 15~20 joints per hour running speed was achieved.

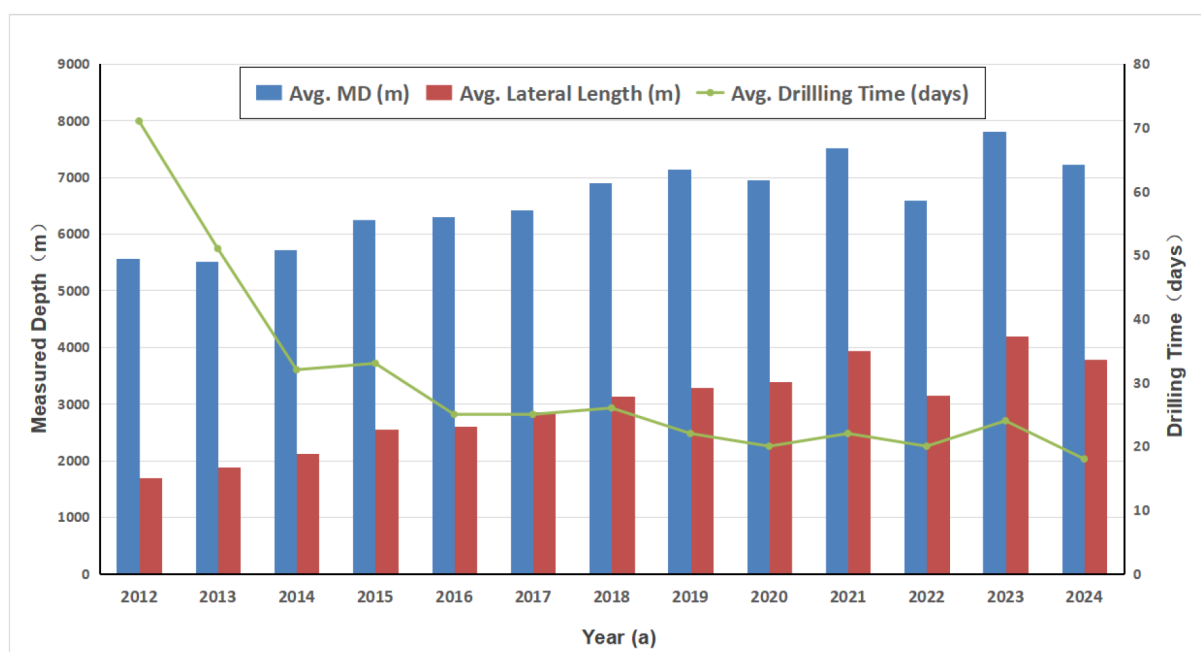


Figure 15—Trend of drilling KPI over years.

## Conclusion

Continuous refinements and innovation have driven consistent growth in average drilling depth—from 5,571m (2012) to 7,210m (2024)—while laterals stretched from 1,702m to 3,890m over the same period. Drilling duration sharply declined from 71 days in 2012 to just 20 days by 2024, coinciding with a 48% reduction in cost per meter drilled. Breakthrough achievements include: extending a 6.75" slim hole single-bit run to 4,875m; installing hybrid 5.5"/4.5" casing strings via CRT tools across 8,157m in one trip; and completing an entire well from spud to TD incident-free in record time (19.2 days). Moreover, six horizontal producers now exceed 4,000m laterals averaging above 50m/hr ROP.

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## References

1. Guoxin Li, Kai Luo, Deqin Shi, 2020. Key technologies, engineering management and important suggestions of shale oil/gas development: Case study of a Duvernay shale project in Western Canada Sedimentary Basin. Presented at Petroleum Exploration and Development, China, August, 2020.
2. Eri Sarmantua Sitinjak. The Longest Ultra slim well Lateral Section in the World With Single Run an Integrated Drilling Technology and Geo-Steering Approach. Presented at Abu Dhabi International Petroleum Exhibition & Conference to be held in Abu Dhabi, UAE, 9–12 November 2020. SPE-202613-MS. SPE/IADC-202162-MS.
3. M. M. Alabed, N. S. Alsuwaidi, J. S. Dugid, et al, 2023. Longest Extended Reach Drilling Well Worldwide drilled in Middle East, Offshore Abu Dhabi, UAE. Presented at the Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, SPE-216326-MS.
4. Greenwood, J. A. Improvements in the Root Cause Analysis of Drillstring Vibration. Presented at Society of Petroleum Engineers. doi:[10.2118/178819-M](https://doi.org/10.2118/178819-M). March 1st, 2016